A Markov-State Heston Extension for Inventory-Aware Market Making

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Abstract

We develop and test a Markov-state extension of a stochastic-volatility market-making model. The baseline Heston formulation captures time-varying variance in the mid-price, while the proposed extension adds a finite-state Markov chain that classifies market regimes using volatility and order-book imbalance. The resulting control problem is a coupled system of HJB equations over time, variance, inventory and regime. On a large Binance replay sample dated 2025-11-17, the Markov-state Heston strategy improves final marked PnL from 25.27 for the constrained-inventory benchmark and 28.63 for the plain Heston model to 41.80. It also reduces maximum absolute inventory from 0.2953 and 0.2799 to 0.1933 and improves the PnL growth-quality score from 0.506 and 0.602 to 0.995. These results suggest that regime-conditioned volatility control can improve both profitability and inventory behavior in high-frequency market making.

Keywords: market making; stochastic control; HJB equation; Heston volatility; Markov states; inventory control; Binance backtesting

1. Motivation and contribution

A market maker earns the spread only if inventory risk is kept under control. A strategy can show positive marked PnL while it simply accumulates inventory and benefits from a favorable terminal mark. Such behavior is not sufficient for a robust market-making model. The goal of this study is therefore twofold: to improve final marked PnL and to improve the shape of the inventory and PnL trajectories.

The proposed model starts from a Heston stochastic-volatility market-making formulation and adds a discrete Markov state variable. The state captures persistent microstructure regimes, such as high-volatility versus low-volatility periods and buy-side versus sell-side book imbalance. The Markov generator couples the value functions across regimes, so the strategy does not solve independent HJB equations per state; instead it solves a single coupled HJB system.

2. Benchmark model: constrained-inventory Guéant-Lehalle formulation

The constrained-inventory benchmark assumes a Brownian reference price and exponential arrival intensities for liquidity-consuming orders. In this model, the HJB can be transformed into a system of linear ordinary differential equations over the inventory grid. This makes it a natural baseline for comparison with stochastic-volatility models.

$$dS_t = \sigma\, dW_t, \qquad \lambda^b(\delta^b) = Ae^{-k\delta^b}, \quad \lambda^a(\delta^a) = Ae^{-k\delta^a}.$$

$$\dot{v}_q(t) = \alpha q^2 v_q(t) - \eta\{v_{q-1}(t) + v_{q+1}(t)\}, \quad v_q(T) = 1,$$

$$\alpha = \frac{k}{2}\gamma\sigma^2, \qquad \eta = A\left(1 + \frac{\gamma}{k}\right)^{-(1 + k/\gamma)}.$$

$$\delta^{b,*}(t,q) = \frac{1}{k}\log\frac{v_q(t)}{v_{q+1}(t)} + \frac{1}{\gamma}\log\left(1 + \frac{\gamma}{k}\right), \quad \delta^{a,*}(t,q) = \frac{1}{k}\log\frac{v_q(t)}{v_{q-1}(t)} + \frac{1}{\gamma}\log\left(1 + \frac{\gamma}{k}\right).$$

The benchmark is useful because it directly penalizes inventory through the ODE coefficients. However, it does not explicitly condition on realized stochastic variance or on market-state regimes.

3. Heston stochastic-volatility market-making model

The Heston model replaces the constant-volatility reference price with a stochastic-variance process. This allows the optimal quote depths to widen or tighten as instantaneous variance changes.

$$dS_t = \mu\,dt + \sqrt{\nu_t}\,dB_t^1 + \varepsilon^+\,dM_t^+ - \varepsilon^-\,dM_t^-, \qquad d\nu_t = \kappa_\nu(\theta - \nu_t)dt + \xi\sqrt{\nu_t}\,dB_t^2.$$

With quadratic utility and running inventory penalty, the objective is written as follows.

$$W(t,x,S,\nu,q) = \sup_{\delta^\pm} \mathrm{E}\left[X_T + q_T S_T - \alpha q_T^2 - \psi\xi^2 \int_t^T q_s^2 \nu_s\,ds\right].$$

The resulting HJB is solved numerically by finite differences on a variance grid and an inventory grid. The quote depths are computed from finite differences of the reservation-value function across adjacent inventory levels.

4. Markov-state Heston extension: mathematical derivation

This section gives the full control construction of the proposed Markov-state Heston model. The aim is to keep the stochastic-volatility structure of the Heston market-making model while allowing all key market-making primitives to depend on a discrete microstructure state. The state is not merely a label used after solving the control problem; it is part of the controlled Markov system and appears directly in the HJB through its generator.

4.1 Market-state process

Let I_t denote the market regime observed by the market maker. It is modeled as a finite-state continuous-time Markov chain with state space S={1,...,K}. The empirical state can be built from volatility buckets, book-imbalance buckets and optional trade-flow buckets. The transition matrix is estimated from bucketed market data and converted into a generator B.

$$I_t \in \mathcal{S} = \{1,\dots,K\}, \qquad \mathbb{P}(I_{t+dt} = j \mid I_t = i) = \mathbf{1}_{i=j} + B_{ij}\,dt + o(dt).$$

$$B_{ij} \geq 0 \ (i \neq j), \qquad B_{ii} = -\sum_{j \neq i} B_{ij}, \qquad \sum_j B_{ij} = 0.$$

$$I_t = \Phi\{\hat{\nu}_t,\ \text{imbalance}_t,\ \text{trade flow}_t\}, \qquad \Phi : \mathbb{R}^d \to \{1,\dots,K\}.$$

The generator B is crucial: it prevents the problem from becoming K unrelated control problems. When the current state is i, the market maker accounts for the possibility that the market may jump to state j during the next infinitesimal interval. This creates a value-transfer term in the HJB.

4.2 State-dependent price, variance and order-flow dynamics

Conditional on I_t=i, the mid-price follows a Heston-type stochastic-volatility dynamics with order-arrival jumps in the mid-price. The variance process is mean-reverting and state-dependent. The correlation between price and variance shocks is also allowed to depend on the state.

$$dS_t = \mu_i\,dt + \sqrt{\nu_t}\,dB_t^{1,i} + \varepsilon_i^+\,dM_t^+ - \varepsilon_i^-\,dM_t^-,$$

$$d\nu_t = \kappa_{\nu,i}(\theta_i - \nu_t)dt + \xi_i\sqrt{\nu_t}\,dB_t^{2,i}, \qquad d\langle B^{1,i}, B^{2,i}\rangle_t = \rho_i\,dt.$$

$$\mathbb{E}[dM_t^+ \mid I_t = i] = \lambda_i^+ \, dt, \qquad \mathbb{E}[dM_t^- \mid I_t = i] = \lambda_i^- \, dt.$$

The parameters mu_i, kappa_{v,i}, theta_i, xi_i, rho_i, lambda_i^+, lambda_i^-, epsilon_i^+ and epsilon_i^- are estimated conditionally on the observed state. In the simplest implementation, only a subset of these parameters is state-dependent for statistical stability; the mathematical formulation allows the full state-dependent specification.

4.3 Quotes, fills, cash and inventory

At time t the market maker posts an ask quote above the mid-price and a bid quote below the mid-price. We use the plus sign for the ask/sell side and the minus sign for the bid/buy side. Fill intensities decay exponentially with the distance from the mid-price.

$$p_t^+ = S_t + \delta_t^+, \quad p_t^- = S_t - \delta_t^-, \qquad \delta_t^+ \geq 0, \ \delta_t^- \geq 0.$$

$$\Lambda_i^+(\delta^+) = \lambda_i^+ e^{-k_i^+ \delta^+}, \qquad \Lambda_i^-(\delta^-) = \lambda_i^- e^{-k_i^- \delta^-}.$$

$$dq_t = dN_t^- - dN_t^+, \qquad dX_t = (S_t + \delta_t^+) dN_t^+ - (S_t - \delta_t^-) dN_t^-.$$

$$q_t \in \mathcal{Q} = \{q_{\min}, q_{\min} + 1, \ldots, q_{\max}\}, \qquad q_{\min} < 0 < q_{\max}.$$

The inventory convention is q=N^- - N^+. Thus a bid fill increases inventory and an ask fill decreases inventory. The inventory is constrained to a finite grid. At the upper boundary the bid side is suppressed because additional buying is forbidden. At the lower boundary the ask side is suppressed because additional selling is forbidden.

4.4 Objective function

The market maker maximizes expected terminal marked wealth subject to a terminal liquidation penalty and a running inventory penalty. The running penalty is proportional to instantaneous variance: holding inventory is more expensive when the market is volatile.

$$W_i(t, x, S, \nu, q) = \sup_{\delta^\pm \in \mathcal{A}} \mathbb{E}_{t,i,x,S,\nu,q}\left[X_T + q_T S_T - \alpha q_T^2 - \psi_i \xi_i^2 \int_t^T q_s^2 \nu_s \, ds\right].$$

The coefficient alpha penalizes leftover terminal inventory. The coefficient psi_i penalizes inventory during the trading session. In practical calibration, psi_i may be fixed across states or allowed to increase in high-volatility states.

4.5 Full HJB equation before reduction

Let L_i^0 be the diffusion generator acting on the variables S and nu when the regime is i. For a smooth test function f it is:

$$\mathcal{L}_i^0 f = \mu_i \partial_S f + \kappa_{v,i}(\theta_i - \nu)\partial_\nu f + \tfrac{1}{2}\nu \partial_{SS} f + \rho_i \xi_i \nu \partial_{S\nu} f + \tfrac{1}{2}\xi_i^2 \nu \partial_{\nu\nu} f.$$

The HJB contains five parts: time evolution, the Heston diffusion generator, the running inventory penalty, the Markov-state switching term, and the two controlled execution Hamiltonians. In compact form:

$$0 = \partial_t W_i + \mathcal{L}_i^0 W_i - \psi_i \xi_i^2 \nu q^2 + \sum_{j=1}^{K} B_{ij}(W_j - W_i)$$

$$+\sup_{\delta^+}\left\{\Lambda_i^+(\delta^+)\Delta_i^{+,\mathrm{fill}}W+(\lambda_i^+-\Lambda_i^+(\delta^+))\Delta_i^{+,\mathrm{nofill}}W\right\}$$

$$+\sup_{\delta^-}\left\{\Lambda_i^-(\delta^-)\Delta_i^{-,\mathrm{fill}}W+(\lambda_i^--\Lambda_i^-(\delta^-))\Delta_i^{-,\mathrm{nofill}}W\right\}.$$

The fill and no-fill increments represent the two possible effects of an incoming market order: it may hit the market maker quote, or it may move the reference price without filling the market maker order. The ask-side and bid-side fill increments are:

$$\Delta_i^{+,\mathrm{fill}}W=W_i(t,x+S+\delta^+,S+\varepsilon_i^+,\nu,q-1)-W_i(t,x,S,\nu,q),$$

$$\Delta_i^{-,\mathrm{fill}}W=W_i(t,x-S+\delta^-,S-\varepsilon_i^-,\nu,q+1)-W_i(t,x,S,\nu,q).$$

The corresponding no-fill increments have the same mid-price jump but leave cash and inventory unchanged. This is the adverse-selection component: even if the market maker is not filled, order flow can move the reference price.

4.6 Linear-quadratic ansatz and reduced reservation equation

The terminal payoff suggests separating cash and marked inventory from the remaining reservation component. We therefore use the following ansatz:

$$W_i(t,x,S,\nu,q)=x+q(S-\alpha q)+u_i(t,\nu,q),\qquad u_i(T,\nu,q)=0.$$

Substitution into the full HJB cancels the cash and linear mid-price terms and leaves a coupled nonlinear equation for u_i(t,nu,q). For interior inventory levels q_min<q<q_max and before optimizing over delta, the equation is:

$$\psi_i\xi_i^2\nu q^2-\mu_i q=\partial_t u_i+\kappa_{\nu,i}(\theta_i-\nu)\partial_\nu u_i+\tfrac{1}{2}\xi_i^2\nu\partial_{\nu\nu}u_i+\sum_j B_{ij}(u_j-u_i)$$

$$+\lambda_i^+\left[q\varepsilon_i^++e^{-k_i^+\delta^+}\{\delta^+-\varepsilon_i^+-\alpha(1-2q)+u_i(q-1)-u_i(q)\}\right]$$

$$+\lambda_i^-\left[-q\varepsilon_i^-+e^{-k_i^-\delta^-}\{\delta^--\varepsilon_i^--\alpha(1+2q)+u_i(q+1)-u_i(q)\}\right].$$

The first line is the state-dependent Heston reservation dynamics. The generator term sum_j B_ij(u_j-u_i) couples all market states. The second and third lines are the ask and bid execution Hamiltonians after the jump and inventory effects have been inserted.

4.7 First-order optimality and optimal quote depths

For the ask side define A_i^+(t,nu,q) as the non-control part of the execution reward. For the bid side define A_i^-(t,nu,q) analogously:

$$A_i^+(t,\nu,q)=\varepsilon_i^++\alpha(1-2q)-u_i(t,\nu,q-1)+u_i(t,\nu,q),$$

$$A_i^-(t,\nu,q) = \varepsilon_i^- + \alpha(1+2q) - u_i(t,\nu,q+1) + u_i(t,\nu,q).$$

Each Hamiltonian has the elementary form exp(-k delta)(delta-A). Differentiating gives the first-order condition:

$$\frac{d}{d\delta}\left[e^{-k\delta}(\delta - A)\right] = e^{-k\delta}\{1 - k(\delta - A)\} = 0 \quad \Longrightarrow \quad \delta^* = A + \frac{1}{k}.$$

Therefore the optimal ask and bid depths in state i are:

$$\delta_i^{+,*} = \frac{1}{k_i^+} + \varepsilon_i^+ + \alpha(1-2q) - u_i(t,\nu,q-1) + u_i(t,\nu,q),$$

$$\delta_i^{-,*} = \frac{1}{k_i^-} + \varepsilon_i^- + \alpha(1+2q) - u_i(t,\nu,q+1) + u_i(t,\nu,q).$$

This formula has an intuitive decomposition. The term 1/k_i is the liquidity-compensation component. The epsilon term compensates for adverse-selection jumps. The alpha term pushes the strategy toward liquidation control. The difference of reservation values u_i(q)-u_i(q±1) is the endogenous inventory skew produced by the HJB.

4.8 Verification equation after inserting the optimal controls

Inserting the optimal depths back into the reduced HJB gives the verification equation that is solved numerically. For interior inventory values:

$$\psi_i \xi_i^2 \nu q^2 - \mu_i q = \partial_t u_i + \kappa_{\nu,i}(\theta_i - \nu)\partial_\nu u_i + \frac{1}{2}\xi_i^2 \nu \partial_{\nu\nu} u_i + \sum_j B_{ij}(u_j - u_i)$$

$$+\lambda_i^+ \left[q\varepsilon_i^+ + \frac{1}{k_i^+}\exp(-1 - k_i^+ A_i^+(t,\nu,q))\right]$$

$$+\lambda_i^- \left[-q\varepsilon_i^- + \frac{1}{k_i^-}\exp(-1 - k_i^- A_i^-(t,\nu,q))\right].$$

This equation is nonlinear because the exponential execution terms contain differences of u across inventory levels. It is also coupled across market states through the generator B. This is the core mathematical difference between the Markov-state Heston extension and a plain Heston model: the latter solves only one variance-inventory surface, while the former solves K interacting surfaces.

4.9 Boundary conditions and admissible controls

At the inventory boundaries one side of the book is removed. If the market maker is at the maximum long inventory, further bid-side buying is not admissible. If the market maker is at the maximum short inventory, further ask-side selling is not admissible.

$$q = q_{\max}: \ \delta_i^{-,*} = +\infty, \qquad q = q_{\min}: \ \delta_i^{+,*} = +\infty.$$

The terminal condition is u_i(T,nu,q)=0 for every state, variance node and inventory level. Neumann-type boundary conditions are used on the variance grid in the finite-difference implementation.

4.10 Numerical solution

The computational grid is indexed by state i, time step m, variance node n, and inventory node r. The value u_i(t,nu,q) is approximated as follows:

$$u_{i,m,n,r} \approx u_i(t_m, \nu_n, q_r), \qquad t_m = m\Delta t, \quad \nu_n = \nu_{\min} + n\Delta\nu.$$

The PDE is solved backward from terminal time. The variance drift is treated with an upwind finite difference, the second variance derivative with a central difference, and the Markov generator with the current state vector at the same time layer. A compact representation of the backward scheme is:

$$\frac{u_{i,m+1,n,r} - u_{i,m,n,r}}{\Delta t} + \mathcal{H}_{i,n,r}(u_{m+1}) = 0, \qquad u_{i,M,n,r} = 0.$$

After the grid is solved, the strategy stores the quote-depth surfaces delta_i^{+,*}(t,nu,q) and delta_i^{-,*}(t,nu,q). During the backtest, the current market state, estimated variance and inventory are mapped to the closest grid location or interpolated in the variance dimension. The resulting model quote is then passed to the execution-aware transformation described next.

5. Execution-aware quote transformation

The theoretical HJB produces optimal offsets under model assumptions. In a discrete limit-order-book backtest, additional execution-layer controls are needed because quote priority, adverse selection and finite tick size may dominate the raw theoretical offset. We use three controls: a flow-balance emission bias, a minimum quote distance, and a push-only inventory buffer.

$$\widetilde{\delta}^{-} = \max(\delta^{-}, \delta_{\min}) + b^{+} + \beta\,[z]_{+}, \qquad \widetilde{\delta}^{+} = \max(\delta^{+}, \delta_{\min}) + b^{-} + \beta\,[-z]_{+},$$

Here z is normalized inventory deviation from the target inventory, beta is the push-only spread buffer, and b+ and b- are the positive and negative parts of the flow-balance bias. The important design choice is push-only inventory control: when inventory is long, the bid is moved farther away instead of pulling the ask closer. This discourages additional buying without selling too cheaply.

6. Test Sample from Binance for 2025-11-17

The main empirical comparison uses the Binance replay file backtest_20251117.parquet. The sample contains 687,916 events and is bucketed at 0.1 seconds for parameter estimation and research replay. All methods use the same inventory grid q in [-400,400], inventory step 0.001, tick size 0.01, zero target inventory, and symmetric negative inventory permission. The maximum inventory range is therefore [-0.4,0.4].

The primary metrics are marked PnL, PnL growth quality, inventory RMS, mean absolute inventory, maximum absolute inventory, terminal inventory, and drawdown normalized by final PnL. PnL and inventory are computed as:

$$\mathrm{PnL}_t = C_t + q_t S_t^{\mathrm{mark}}, \quad I_{\mathrm{RMS}} = \sqrt{T^{-1}\int_0^T q_t^2 dt}, \quad Q_{\mathrm{PnL}} = \frac{\overline{\Delta PnL}}{\mathrm{sd}(\Delta PnL)}\sqrt{N}.$$

7. Empirical results on Test Sample from Binance for 2025-11-17

Model	Final marked PnL	PnL quality	Max |inventory|	Inventory RMS	Terminal inventory	Drawdown / final PnL
Guéant-Lehalle	25.2667	0.5064	0.295320	0.129133	0.286240	1.2896
Heston	28.6267	0.6022	0.279940	0.123435	0.270860	1.0731
Markov-state Heston	41.7954	0.9954	0.193340	0.107204	0.107890	0.9466

Table 1. Comparative metrics on the Test Sample from Binance for 2025-11-17.

The Markov-state Heston strategy achieves the largest final marked PnL, with 41.80 versus 28.63 for the plain Heston model and 25.27 for the constrained-inventory benchmark. Importantly, this improvement is not obtained by letting the inventory hit the boundary. Maximum absolute inventory decreases to 0.1933, compared with 0.2799 for Heston and 0.2953 for the benchmark. Terminal inventory is also materially smaller.

The PnL-growth quality score, defined as the average PnL increment divided by its standard deviation and scaled by the square root of the number of increments, rises to 0.995. This indicates that the PnL curve is less dominated by noisy inventory swings and closer to a stable upward trajectory in increment-quality terms.

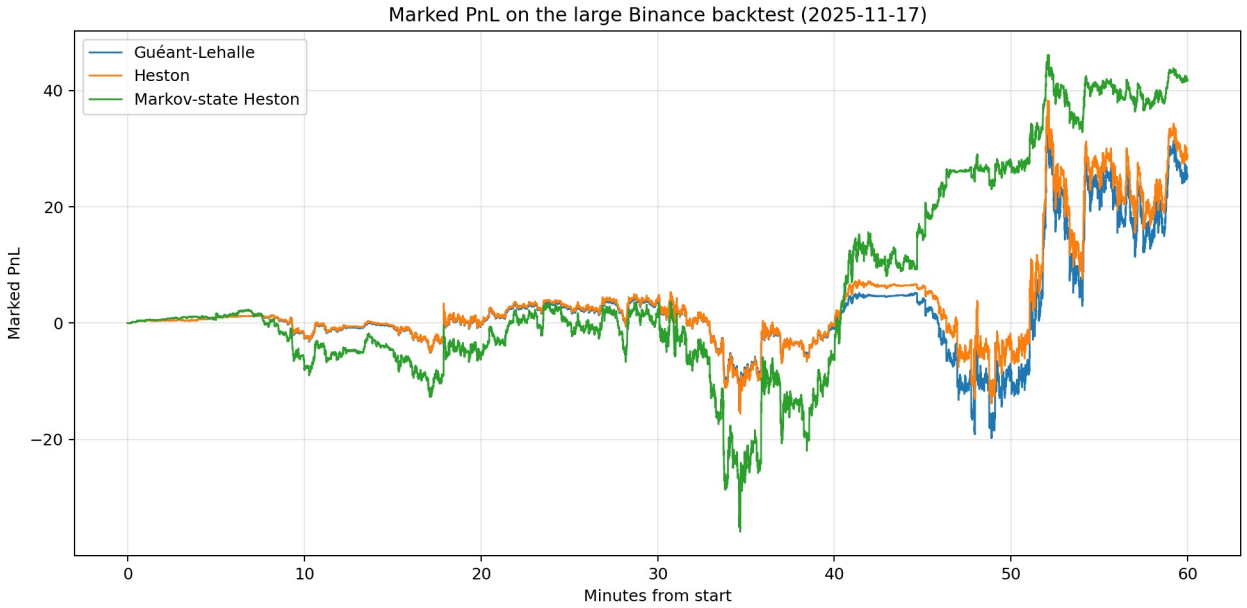


Figure 1. Marked PnL curves on the Test Sample from Binance for 2025-11-17.

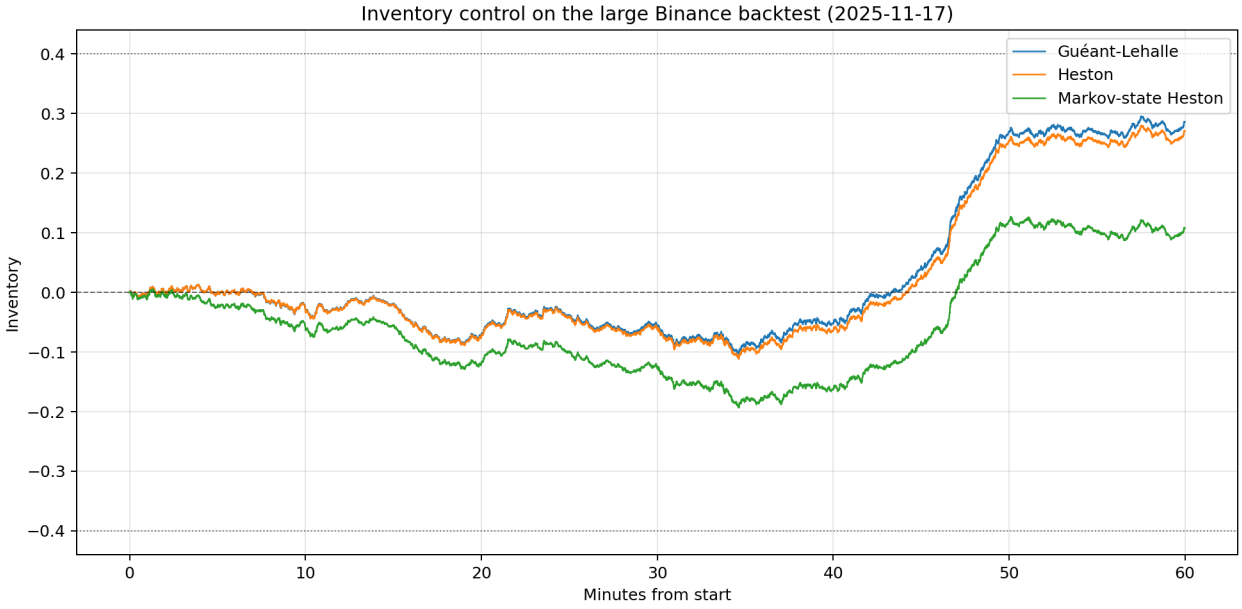


Figure 2. Inventory trajectories. The proposed model avoids persistent boundary inventory and ends closer to zero.

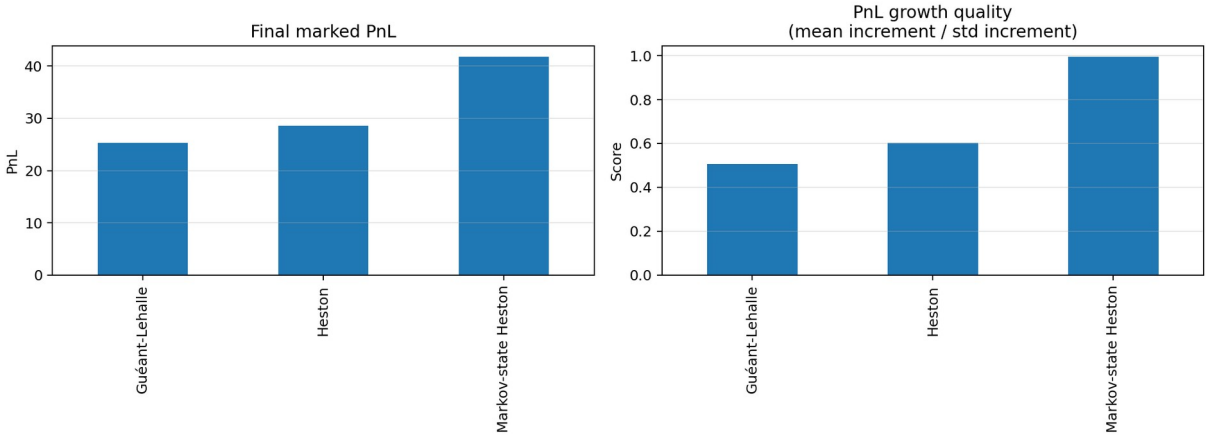


Figure 3. Final marked PnL and PnL growth-quality score.

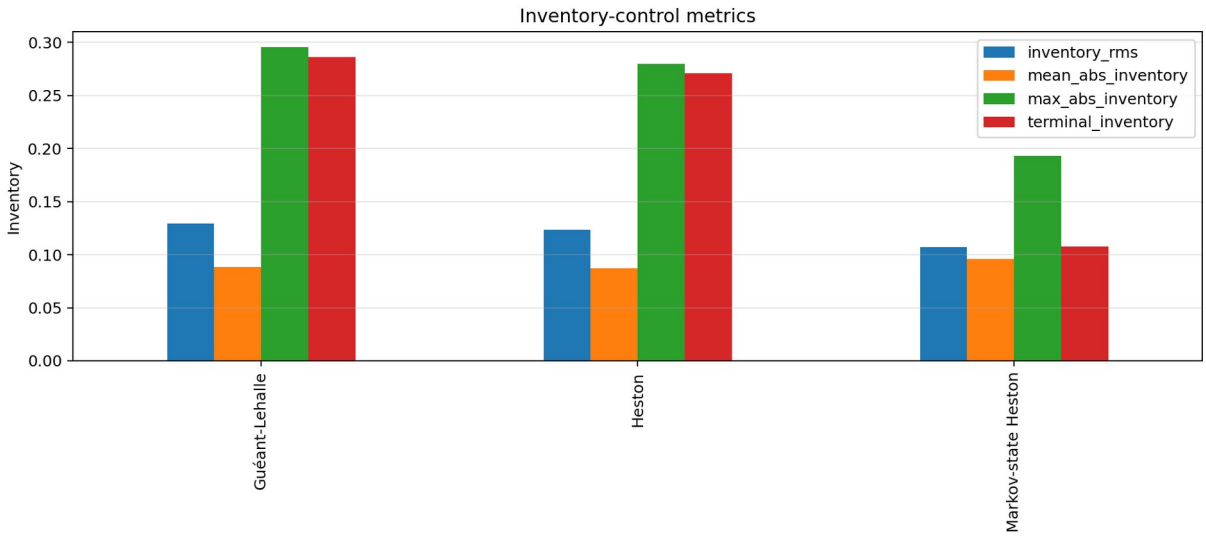


Figure 4. Inventory-control metrics: RMS, mean absolute inventory, maximum absolute inventory and terminal inventory.

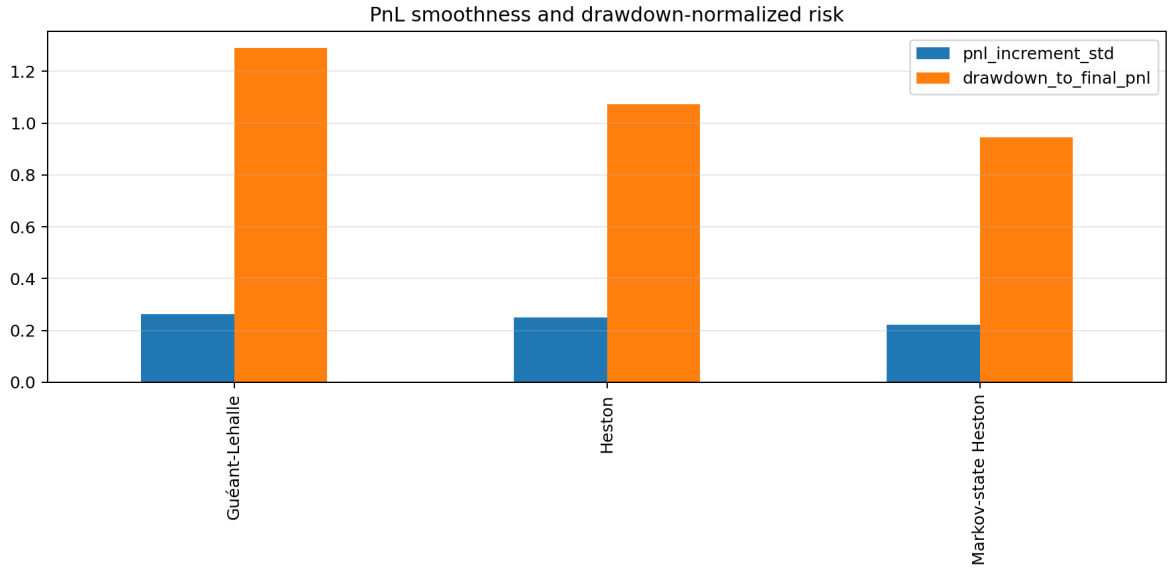


Figure 5. PnL smoothness and normalized drawdown diagnostics.

8. Interpretation

The plain Heston model improves the benchmark in final marked PnL because stochastic variance allows the strategy to adapt quote depths to changing volatility. The Markov-state Heston model improves further because it separates market regimes that have different execution and inventory risks. When the market enters a state with stronger imbalance or different volatility, the value surface and optimal quote depths change accordingly.

The inventory result is central. A pure directional strategy can produce marked PnL by accumulating inventory. The proposed model shows a better inventory profile: lower maximum absolute inventory, lower RMS inventory than the alternatives, and materially lower terminal inventory. This is closer to the desired market-making behavior: earning spread while keeping exposure controlled.

9. Limitations and future validation

The current empirical evidence is based on historical Binance data for 2025-11-17. The analysis is therefore limited to one market venue and one trading date, even though it provides a useful first comparison of PnL shape, inventory stability and state-conditioned quote control. Future work will extend the same methodology to Moscow Exchange data, including instruments with different tick sizes, liquidity profiles and queue dynamics. We also plan controlled live or paper-trading runs to evaluate real-time operational stability, latency, exchange fees and order-priority effects. Additional out-of-sample dates should be used to test parameter robustness and the persistence of the inventory-control improvement.

10. Conclusion

The Markov-state Heston extension provides a practical way to combine stochastic-volatility market making with regime-dependent microstructure information. On the Test Sample from Binance for 2025-11-17, it improves final marked PnL, reduces inventory boundary behavior, and improves PnL growth quality relative to both the constrained-inventory benchmark and the plain Heston model. The result supports the central thesis: stochastic volatility is valuable, but conditioning the HJB on Markov market states materially improves the trading profile.

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